

EcoSem-VQA: Green Cross-Layer Semantic Communication for Edge Visual Question Answering via Joint Rate and Token Budgeting

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Abstract

Deploying edge-assisted Visual Question Answering (VQA) over resource-constrained wireless links requires balancing physical-layer transmission energy against server-side model inference complexity. Existing semantic communication (SemCom) frameworks predominantly presume analog, unquantized transmissions, employ toy-scale decoders, and overlook the energy consumption of large foundational Vision–Language Models (VLMs). This paper proposes **EcoSem-VQA**, an energy-frugal, cross-layer task-oriented semantic communication system for edge VQA under standardized digital physical layers. EcoSem-VQA integrates deep variational neural image compression, a digital-compatible physical layer (**DigiSem-PHY**) featuring 3GPP 5G-compliant LDPC(1020, 512) coding over Rician fading ($K = 6$ dB), and an edge-deployed Qwen3-VL-4B model adapted via parameter-efficient low-rank adaptation (LoRA). Under a rigorous dual-mode green energy formulation comparing Constant Symbol-Power Mode (CSPM) and Constant Query-Energy Mode (CQEM), we demonstrate that CQEM activates a Physical Power Concentration (PPC) mechanism: selecting compact source bitrates unlocks up to +3.01 dB and +6.00 dB of physical SNR gain, shielding the link from catastrophic packet erasure. We develop a lightweight Channel-Aware Rate and Token Budgeting Network (**CART-Net**) that inspects multimodal query semantics and instantaneous physical channel state information (CSI) to dynamically co-optimize transmitter source bitrates (2k, 4k, 8k bytes) and a receiver Dynamic Token Budget Allocator (**DyTBA**, $T_v \in \{\text{low, medium, high}\}$). Extensive evaluation

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over 2,400 independent blind test images across 10 channel SNR points (-5.0 dB $\dots$ 20.0 dB) on the TDIUC benchmark demonstrates that CART-Net attains 36.92% accuracy under harsh fading (2.5 dB), outperforming fixed 4k by +29.92% and channel-blind policies by +27.71%. Furthermore, DyTBA autonomously prunes visual tokens by 49.2% for low-entropy semantic tasks while strictly defining the non-dominated Pareto efficiency frontier across RF energy, system latency, and task accuracy.

Index Terms

Green communications, semantic communications, visual question answering, vision–language models, cross-layer optimization, LDPC codes, energy efficiency.

I. INTRODUCTION

THE proliferation of edge intelligence and multimodal foundation models has spurred the deployment of edge-assisted Visual Question Answering (VQA) in mission-critical Internet-of-Things (IoT) and cyber-physical networks, including autonomous unmanned aerial vehicles (UAVs), remote environmental sensing, and robotic search-and-rescue [1], [2]. In these wireless scenarios, a battery-powered edge sensor captures high-resolution imagery and offloads visual data over an unreliable, fading wireless uplink to an edge computing node hosting a multi-billion-parameter Vision–Language Model (VLM) [3], [4]. The edge VLM executes joint visual perception and natural language reasoning to respond to complex semantic queries in real time.

However, delivering high-accuracy VQA services over wireless networks encounters two severe, coupled bottlenecks:

- 1) **Wireless Fading Outages and Channel Degradation:** Wireless channels are inherently subject to multipath fading, shadowing, and path loss. Under classical separate source-channel coding (SSCC) and block-length constrained digital codes such as low-density parity-check (LDPC) codes, channel attenuation triggers the notorious “cliff effect” [5]. When channel conditions drop below the waterfall threshold, block decoding fails entirely, corrupting the bitstream and leading to catastrophic answer errors.
- 2) **Multimodal Token Explosion and Edge Compute Costs:** Contemporary autoregressive VLMs divide input images into a fine-grained grid of visual patches, projecting them into lengthy sequences of visual tokens [3]. The computational complexity of the Transformer self-attention mechanism scales quadratically with the visual sequence length, $\mathcal{O}(T_v^2)$ [18]. For simple, coarse-grained semantic

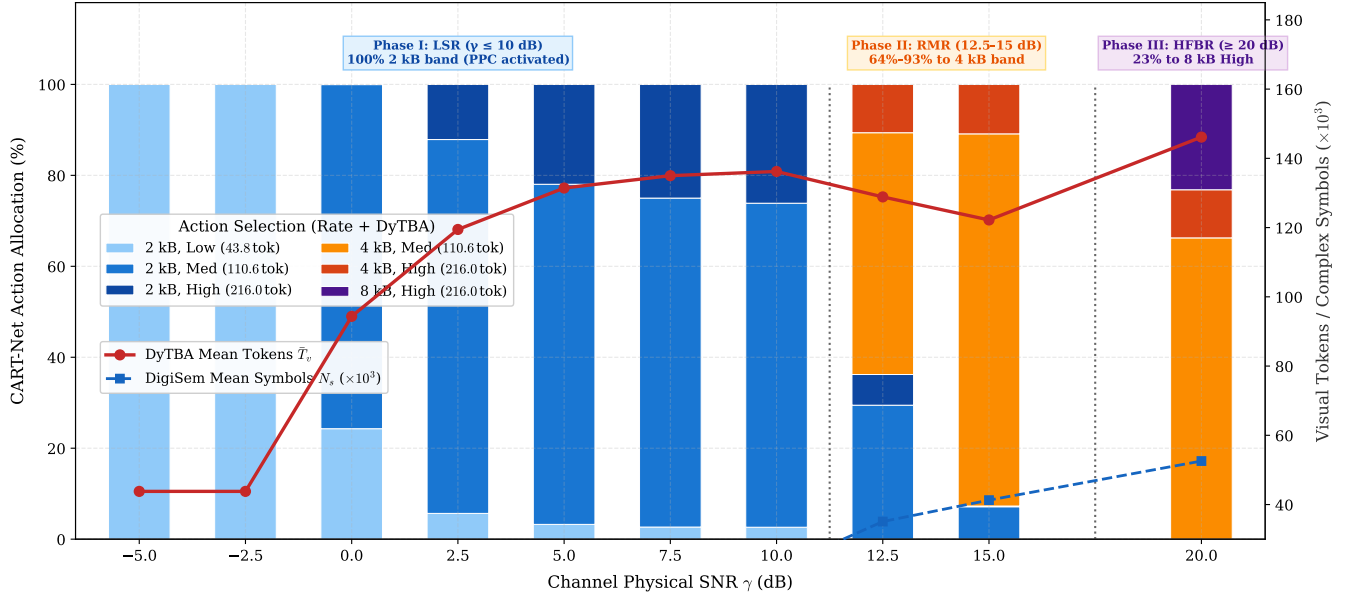


Fig. 1. System framework and dynamic action routing of the proposed EcoSem-VQA framework. (a) A lightweight transmitter-side CART-Net policy inspects question text embeddings, an image spatial preview, and physical channel state information (CSI) γ , dynamically selecting an action $a \in \mathcal{A}$ spanning 3 source bitrates and 3 DyTBA receiver visual token budgets. (b) Empirical routing distribution across 2,400 blind test images over Rician fading ($K = 6$ dB), illustrating the emergent three-phase adaptation mechanics: Link Survival Regime (LSR, ≤ 10 dB), Rate Migration Regime (RMR, 12.5 $\sim$ 15 dB), and High-Fidelity Burst Regime (HFBR, ≥ 20 dB).

queries (e.g., verifying object presence), processing hundreds of redundant visual tokens imposes exorbitant computational and energy overhead on the edge server without tangible task accuracy improvement.

To break through the Shannon capacity bottleneck, *semantic communication* (SemCom) has emerged as a cornerstone of next-generation 6G communications [6], [7]. Rather than seeking error-free bit reconstruction, task-oriented semantic communication extracts and conveys only task-relevant semantics. For multimodal reasoning, pioneer frameworks such as MU-DeepSC [8], [9] and Goal-Oriented Scene-Graph communication [10] have successfully demonstrated the power of joint source-channel transceivers.

Nevertheless, translating existing semantic communication theories into practical wireless networks exposes critical design deficiencies:

- **Reliance on Unquantized Analog Transmission:** The vast majority of existing SemCom paradigms (e.g., DeepJSCC and MU-DeepSC) rely on analog transmission, mapping latent neural features directly into continuous channel symbols without digital constellation mapping or error-correction framing. This assumption conflicts with standardized digital communication infrastructure (e.g., 3GPP LDPC, QPSK/16-QAM) and overlooks discrete packet erasure dynamics.
- **Neglect of Downstream Edge Inference Energy:** Existing works predominantly focus on reducing radio transmission bandwidth while evaluating toy-scale classification decoders (e.g., MCAN or 3-

layer CNNs). In practical edge systems running 4B-to-7B parameter VLMs, the energy consumed by neural inference equals or surpasses radio transmission energy [2], [19].

- **Rigid Physical Energy Trade-offs:** Conventional systems enforce static transmission parameters. In energy-constrained edge uplinks (e.g., battery-operated aerial sensors), transmitting a compact source payload over fewer channel symbols enables the transmitter to allocate higher energy per symbol. This unlocks a significant effective signal-to-noise ratio (SNR) boost that can shield the communication link from deep fading outages.

To address these challenges, this paper presents **EcoSem-VQA**, an end-to-end, cross-layer task-oriented semantic communication framework for edge VQA under realistic digital channel coding and Rician fading. As depicted in Fig. 1, the system features a deep variational neural image compression codec, a digital-compatible physical layer (**DigiSem-PHY**) with 3GPP 5G-compliant LDPC(1020, 512) coding, and an edge-deployed Qwen3-VL-4B foundation model adapted via low-rank adaptation (LoRA) [11], [12]. A lightweight Channel-Aware Rate and Token Budgeting Network (**CART-Net**) dynamically co-optimizes the source bitrate $B \in \{2k, 4k, 8k\}$ bytes and the receiver Dynamic Token Budget Allocator (**DyTBA**, $T_v \in \{\text{low}, \text{medium}, \text{high}\}$).

The primary contributions of this paper are summarized as follows:

- 1) **Digital-Compatible Cross-Layer SemCom Architecture (EcoSem-VQA):** We propose an integrated semantic communication framework that couples deep variational source compression with standardized 3GPP digital channel coding (LDPC(1020, 512), DigiSem-PHY) and a foundational VLM (Qwen3-VL-4B). This establishes a practical, digitally deployable semantic pipeline.
- 2) **Dual-Mode Energy Formulation and Physical Power Concentration (PPC):** We establish rigorous mathematical energy models encompassing both wireless transmission and edge VLM inference, formulating two operational modes: Constant Symbol-Power Mode (CSPM) and Constant Query-Energy Mode (CQEM). We prove that under CQEM, transmitting a compact 2000-byte payload activates a physical power concentration (PPC) mechanism, yielding an exact +3.01 dB (vs. 4000 bytes) and +6.00 dB (vs. 8000 bytes) physical SNR boost that circumvents the LDPC outage waterfall under harsh fading.
- 3) **Channel-Aware Semantic Policy Network (CART-Net):** We design a lightweight multi-branch policy network mapping query semantic embeddings, image preview features, and instantaneous channel SNR into a 9-action discrete space. CART-Net exhibits an emergent three-phase adaptation:

a *Link Survival Regime* (LSR, ≤ 10 dB), a *Rate Migration Regime* (RMR, $12.5 \sim 15$ dB), and a *High-Fidelity Burst Regime* (HFBR, ≥ 20 dB).

- 4) **Extensive Empirical Validation on TDIUC:** We evaluate EcoSem-VQA over 2,400 independent blind test images across 10 channel SNRs (-5.0 dB $\dots$ 20.0 dB) under Rician fading ($K = 6$ dB). Experimental results show that under harsh fading (2.5 dB, CQEM), CART-Net achieves 36.92% accuracy, delivering a +29.92% advantage over fixed 4k and a +27.71% advantage over channel-blind routing. Furthermore, DyTBA autonomously cuts visual tokens by 49.2% for low-entropy tasks while establishing the non-dominated Pareto frontier across RF energy, system latency, and task accuracy.

The rest of this paper is structured as follows. Section II discusses related work. Section III presents the system model and cross-layer optimization problem. Section IV describes the policy network architecture and its three-phase adaptation. Section V provides extensive empirical evaluations and discussions. Finally, Section VI concludes the paper.

II. RELATED WORK

A. Task-Oriented and Multimodal Semantic Communications

Semantic communication breaks through the Shannon paradigm of bit-level exact reproduction by transmitting only the semantic information essential for downstream task completion [6], [7]. Deep joint source-channel coding (DeepJSCC) demonstrated that neural autoencoders achieve graceful degradation over fading channels, mitigating the cliff effect of separate source-channel coding (SSCC) [5], [13]. In the text and speech domains, DeepSC [14] and DeepSC-S [15] applied Transformer encoders to recover semantic meaning under severe channel fading.

For multimodal tasks, semantic communication has recently expanded into Visual Question Answering (VQA). Xie *et al.* [8], [9] proposed MU-DeepSC, employing cross-attention networks to extract and fuse correlated visual and textual features for edge VQA over Rayleigh channels. Liu *et al.* [10] developed goal-oriented scene-graph transmission, prioritizing bounding boxes and spatial relationships based on query relevance. Similarly, Chappuis *et al.* [16] explored symbolic prompt generation for remote sensing VQA.

Despite their conceptual promise, existing multimodal semantic communication systems rely heavily on two simplifying assumptions:

- 1) *Continuous Analog Symbol Transmission*: Prior works predominantly transmit continuous latent feature vectors directly over AWGN or Rayleigh channels, assuming unquantized, analog physical layers. This precludes direct integration into standardized digital cellular and Wi-Fi architectures governed by discrete constellations (e.g., QPSK, 16-QAM) and block error-correction codes (e.g., LDPC, Polar codes).
- 2) *Toy-Scale Answering Decoders*: Existing systems typically employ small, custom-trained attention networks (such as MCAN or 3-layer Transformers) with limited parameter capacity. In contrast, modern practical VQA tasks rely on foundational Vision–Language Models capable of complex zero-shot reasoning. Adapting semantic communication to real-world, billion-parameter VLMs remains fundamentally unaddressed.

B. Vision–Language Models and Visual Token Management

Recent breakthroughs in multimodal foundation models, such as LLaVA [17] and the Qwen-VL family [3], [4], have unified vision and natural language processing through vision encoders (e.g., ViT) followed by large autoregressive language decoders. These models process visual inputs by dividing images into 14×14 or 28×28 patches, generating a 2D grid of visual tokens that are concatenated with textual query tokens.

While foundational VLMs exhibit superior perceptual and reasoning accuracy, their computational cost is predominantly dominated by the self-attention layers within the language decoder, whose memory and computational complexity scale quadratically with the total token sequence length:

$$\mathcal{C}_{\text{attn}} \propto \mathcal{O}((T_q + T_v)^2 \cdot d), \quad (1)$$

where T_q is the query token length, T_v is the visual token sequence length, and d is the hidden embedding dimension. When $T_v \gg T_q$, the visual tokens account for over 85% of the total inference latency and GPU energy consumption [18].

Although image resizing and visual token pruning have been investigated in standalone computer vision literature to accelerate inference, their interaction with wireless channel conditions and transmission source bitrates has been overlooked. Specifically, how an edge receiver should dynamically scale its visual token budget in tandem with physical layer channel quality and transmitter source coding rates remains an open problem.

C. Green Communications and Cross-Layer Edge Computing

The energy consumption of mobile edge networks has become a central challenge in 6G green communications [2], [19]. Traditional cross-layer resource allocation frameworks coordinate physical layer parameters (such as transmit power, modulation and coding schemes (MCS), and bandwidth) to maximize throughput or spectral efficiency under physical constraints [20].

In the era of edge artificial intelligence, edge optimization must jointly balance radio transmission energy with server-side computation energy [18], [21]. When an edge device operates under a strict battery or energy budget (such as an aerial UAV or battery-powered IoT node), the energy consumed to transmit each query is constrained by a finite quota:

$$E_{\text{tx}} = P_s \cdot N_s \leq E_0, \quad (2)$$

where P_s is the average energy per transmitted complex symbol and N_s is the total number of channel symbols. Under this strict budget constraint, reducing the source bitrate (i.e., decreasing N_s) directly unlocks a physical layer power boost ($\Delta P_s > 0$), resulting in higher effective channel SNR γ_{eff} that provides physical-layer protection against severe fading.

To the best of our knowledge, this work is the first to establish a unified, cross-layer semantic communication framework that co-optimizes the transmitter's neural source compression bitrate, physical LDPC blocklength, and the receiver's VLM visual token budget under realistic Rician fading channels, explicitly establishing the green Pareto efficiency frontier for multimodal edge intelligence.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Overview

We consider a task-oriented wireless edge VQA architecture, termed **EcoSem-VQA**, where a remote wireless edge transmitter (e.g., an aerial UAV or IoT vision sensor) captures an RGB image $I \in \mathbb{R}^{H \times W \times 3}$ and receives a user natural language query q . An edge server equipped with a parameter-efficient fine-tuned Vision–Language Model (VLM) executes multimodal reasoning to infer the ground-truth answer A^* .

To minimize energy consumption while guaranteeing task accuracy over an unreliable wireless link, the system dynamically co-optimizes the source transmission bitrate B and the receiver visual token budget T_v via a Channel-Aware Rate and Token Budgeting Network (**CART-Net**), denoted by π . The overall

pipeline comprises three core subsystems: neural variational source compression, a digital physical layer (**DigiSem-PHY**) with LDPC channel coding over Rician fading, and an edge VLM equipped with a Dynamic Token Budget Allocator (**DyTBA**).

B. Neural Source Compression and DigiSem-PHY Digital Channel Model

1) *Variational Source Compression*: The transmitter employs a deep variational autoencoder with a scale hyperprior architecture [22], [23]. The raw image I is mapped to a latent semantic representation $\mathbf{y} = g_a(I; \theta_a)$ via a convolutional analysis transform. To capture spatial dependencies, a hyper-analysis transform extracts hyper-latents $\mathbf{z} = h_a(\mathbf{y}; \phi_a)$, which parameterize the conditional Gaussian entropy model $p(\tilde{\mathbf{y}} | \hat{\mathbf{z}}) \sim \mathcal{N}(\mathbf{0}, \sigma^2)$ where $\sigma = h_s(\hat{\mathbf{z}}; \phi_s)$.

The quantized latents $\hat{\mathbf{y}} = \lfloor \mathbf{y} \rfloor$ and hyper-latents $\hat{\mathbf{z}} = \lfloor \mathbf{z} \rfloor$ are losslessly encoded into a discrete binary bitstream using an asymmetric numeral systems (ANS) entropy coder. By regulating entropy thresholds, the transmitter controls the total compressed payload size $B \in \mathcal{B} = \{2000, 4000, 8000\}$ bytes.

2) *3GPP LDPC Channel Coding and Digital Modulation*: Unlike prior analog JSCC literature, our physical layer, designated **DigiSem-PHY**, strictly adheres to standardized digital transmission specifications. The binary bitstream of length $8B$ bits is partitioned into N_{blocks} codewords. Each block is appended with a 32-bit cyclic redundancy check (CRC-32) and encoded using a 3GPP 5G-compliant LDPC(N_b, K_b) code, where the codeword length is $N_b = 1020$ bits, the information payload per block is $K_b = 512$ bits (including CRC), and the code rate is $R_c = 512/1020 \approx 0.502$.

Each block of 480 payload bytes (3840 bits) maps into $\lceil 3840/480 \rceil = 8$ LDPC codeblocks. Hence, the total number of LDPC blocks required to transmit B bytes is:

$$N_{\text{blocks}}(B) = \left\lceil \frac{B}{48} \right\rceil. \quad (3)$$

The encoded bits are modulated via Gray-mapped Quadrature Phase Shift Keying (QPSK, modulation order $m = 2$ bits/symbol), producing N_s complex channel symbols:

$$N_s(B) = N_{\text{blocks}}(B) \cdot \frac{N_b}{m} = \left\lceil \frac{B}{48} \right\rceil \cdot 510. \quad (4)$$

Specifically, for our three candidate bitrates:

- $B = 2000 \text{ B} \implies N_{\text{blocks}} = 42 \implies N_s = 21,420 \text{ symbols.}$
- $B = 4000 \text{ B} \implies N_{\text{blocks}} = 84 \implies N_s = 42,840 \text{ symbols.}$

- $B = 8000 \text{ B} \implies N_{\text{blocks}} = 167 \implies N_s = 85,170 \text{ symbols.}$

3) *Rician Fading Channel and Iterative Decoding:* The transmitted symbol vector $\mathbf{x} \in \mathbb{C}^{N_s}$ with $\mathbb{E}[|x_k|^2] = 1$ traverses a block flat-fading Rician channel:

$$y_k = \sqrt{P_s} h_k x_k + n_k, \quad k = 1, \dots, N_s, \quad (5)$$

where P_s is the average transmit energy per symbol, $n_k \sim \mathcal{CN}(0, \sigma_n^2)$ represents complex additive white Gaussian noise, and h_k denotes the Rician fading coefficient characterized by the Rician K -factor:

$$h_k = \sqrt{\frac{K}{K+1}} h_{\text{LOS}} + \sqrt{\frac{1}{K+1}} h_{\text{NLOS}}, \quad (6)$$

where $h_{\text{LOS}} = e^{j\theta}$ is the deterministic line-of-sight (LOS) component, $h_{\text{NLOS}} \sim \mathcal{CN}(0, 1)$ is the scattered Rayleigh non-line-of-sight (NLOS) component, and $K = 6 \text{ dB}$ reflects typical elevated air-to-ground or micro-cellular propagation. The channel gain is normalized such that $\mathbb{E}[|h_k|^2] = 1$. The nominal received SNR is defined as $\gamma = P_s / \sigma_n^2$.

At the receiver, log-likelihood ratios (LLRs) are computed from the channel observations y_k and fed into an iterative Soft Belief Propagation (BP) decoder executing up to 50 iterations with early stopping upon syndrome check satisfaction. A transmission block is declared successful if and only if its CRC-32 checksum passes. The packet delivery ratio (PDR) $\Pi(\gamma, B)$ is the probability that all $N_{\text{blocks}}(B)$ codewords are successfully decoded:

$$\Pi(\gamma, B) = \prod_{i=1}^{N_{\text{blocks}}(B)} (1 - \text{BLER}_i(\gamma)). \quad (7)$$

If any block fails CRC verification, a packet outage occurs, and the image is dropped ($\hat{I} = \emptyset$). Otherwise, the bitstream is assembled and passed to the synthesis transform $g_s(\hat{\mathbf{y}}; \boldsymbol{\theta}_s)$ to reconstruct the RGB image $\hat{I}$.

C. Receiver VLM and Dynamic Token Budget Allocator (DyTBA)

The edge server runs Qwen3-VL-4B-Instruct [3], [4], quantized to 4-bit NormalFloat (NF4) and adapted via parameter-efficient Low-Rank Adaptation (LoRA) on the query, key, value, and output projection layers of the language backbone.

Before feeding the reconstructed image $\hat{I}$ into the Vision Transformer (ViT) encoder, the receiver employs a Dynamic Token Budget Allocator (**DyTBA**) to dynamically resize and tokenize the image

according to a prescribed visual token budget $T_v \in \mathcal{T} = \{\text{low}, \text{medium}, \text{high}\}$:

- **Low Budget** ($T_v \approx 43.6$ tokens): Downsamples the input resolution to an area of 50,176 pixels, capturing global semantic layout with minimal compute.
- **Medium Budget** ($T_v \approx 110.2$ tokens): Scales the input to 100,352 pixels, providing balanced spatial granularity.
- **High Budget** ($T_v \approx 215.8$ tokens): Preserves dense resolution (200,704 pixels) for fine-grained spatial and count reasoning.

The VLM autoregressively generates the predicted answer $\hat{A} = \text{VLM}(\hat{I}, q; T_v)$. Task accuracy is measured by strict string exact match after canonicalization:

$$\text{Acc} = \mathbb{I}(\hat{A} = A^*). \quad (8)$$

D. Energy and Latency Accounting Models

1) *Transmission Power and Energy Modes: CSPM vs. CQEM*: We consider two operational energy modes reflecting distinct real-world deployment constraints:

- **Constant Symbol-Power Mode (CSPM)**: In grid-powered or macro-cell systems, the transmit power per symbol is held constant ($P_s = P_0$). The total RF transmission energy per query scales proportionally with the symbol count:

$$E_{\text{tx}}^{\text{CSPM}}(B) = P_0 \cdot N_s(B). \quad (9)$$

- **Constant Query-Energy Mode (CQEM)**: In battery-constrained edge devices (e.g., UAVs with fixed per-frame transmission energy quotas), the total energy allocated to transmit an image query is clamped at a constant budget E_0 . Consequently, the average transmit energy per symbol adjusts inversely with the symbol count:

$$P_s(B) = \frac{E_0}{N_s(B)}. \quad (10)$$

Under CQEM, selecting a more compact source bitrate $B_1 < B_2$ directly activates the **Physical Power Concentration (PPC)** mechanism, yielding an effective SNR boost:

$$\Delta\gamma_{\text{eff}}(B_1, B_2) = 10 \log_{10} \left(\frac{N_s(B_2)}{N_s(B_1)} \right) \text{ dB}. \quad (11)$$

Specifically, transmitting 2000 bytes ($N_s = 21,420$) instead of 4000 bytes ($N_s = 42,840$) confers an exact:

$$\Delta\gamma = 10 \log_{10} \left(\frac{42,840}{21,420} \right) = +3.01 \text{ dB} \quad (12)$$

power advantage. Compressing from 8000 bytes ($N_s = 85,170$) to 2000 bytes yields a +6.00 dB gain. In deep fading regimes, this +3.01 $\sim$ +6.00 dB PPC boost elevates the operating SNR past the LDPC waterfall threshold, shielding the link from packet erasure.

2) *Edge Computation Energy*: The server-side computation energy consumed by the edge VLM during inference is modeled as:

$$E_{\text{rx,comp}}(T_v) = P_{\text{GPU}} \cdot t_{\text{vlm}}(T_v), \quad (13)$$

where P_{GPU} is the average GPU running power and $t_{\text{vlm}}(T_v)$ is the measured forward inference time, which scales monotonically with the visual token count T_v . The total system energy per query is:

$$E_{\text{total}} = E_{\text{tx}} + E_{\text{rx,comp}}. \quad (14)$$

3) *End-to-End Latency Model*: The total end-to-end processing and communication latency t_{e2e} is given by:

$$t_{\text{e2e}} = t_{\text{enc}}(B) + t_{\text{tx}}(B) + t_{\text{dec}}(B) + t_{\text{vlm}}(T_v), \quad (15)$$

where t_{enc} is the onboard neural source encoding time, $t_{\text{tx}} = N_s(B)/W_{\text{sym}}$ is the wireless airtime over a symbol transmission bandwidth W_{sym} , t_{dec} is the receiver decompression time, and t_{vlm} is the VLM answer generation time.

E. Cross-Layer Optimization Problem

Let $\mathcal{A} = \mathcal{B} \times \mathcal{T}$ denote the set of $|\mathcal{A}| = 9$ joint source-rate and visual-budget action configurations:

$$a = (B, T_v) \in \mathcal{A}. \quad (16)$$

Our objective is to find an optimal cross-layer policy $\pi^*(I, q, \gamma)$ that maximizes the expected end-to-end VQA accuracy while satisfying strict energy and latency constraints:

$$\max_{\pi} \quad \mathbb{E}_{(I, q, \gamma)} \left[\mathbb{I}(\hat{A}(a) = A^*) \cdot \Pi(\gamma_{\text{eff}}(B), B) \right] \quad (17a)$$

$$\text{s.t.} \quad E_{\text{total}}(a) \leq E_{\text{budget}}, \quad \forall(I, q, \gamma), \quad (17b)$$

$$t_{\text{e2e}}(a) \leq \tau_{\text{deadline}}, \quad \forall(I, q, \gamma), \quad (17c)$$

$$a = \pi(I, q, \gamma) \in \mathcal{A}. \quad (17d)$$

Problem (17) is a combinatorial, non-convex stochastic optimization problem involving non-differentiable wireless packet erasure and large-scale multimodal neural reasoning. In the next section, we design an efficient neural policy network to solve it.

IV. CROSS-LAYER SEMANTIC POLICY DESIGN: CART-NET

A. CART-Net Architecture

To resolve the non-convex optimization problem formulated in (17) in real time without imposing heavy computation at the transmitter, we design a lightweight Channel-Aware Rate and Token Budgeting Network, termed **CART-Net** ($\pi_{\theta}(a \mid I, q, \gamma)$). CART-Net jointly inspects multimodal query semantics and instantaneous physical channel quality to output a categorical probability distribution over the $|\mathcal{A}| = 9$ discrete actions:

$$\mathcal{A} = \{2k, 4k, 8k\} \times \{\text{low}, \text{med}, \text{high}\}. \quad (18)$$

As illustrated in Fig. 1(a), CART-Net comprises three input branches:

- 1) **Question Semantic Feature** ($f_q \in \mathbb{R}^{d_q}$): A frozen text embedding (dimension $d_q = 512$) extracts syntactic and semantic intent, distinguishing low-entropy categorical queries (e.g., “Is there a car?”) from high-entropy spatial queries (e.g., “How many pedestrians are to the left of the bench?”).
- 2) **Image Preview Feature** ($f_I \in \mathbb{R}^{d_I}$): A lightweight convolutional feature map (dimension $d_I = 256$) extracted from a sub-sampled preview of the source image I . This preview incurs negligible onboard computational overhead (< 1.5 ms) and captures global visual complexity, object clutter, and texture density.
- 3) **Physical CSI Feature** ($\gamma \in \mathbb{R}$): The instantaneous channel SNR $\gamma = P_s/\sigma_n^2$ estimated by the receiver and fed back via a low-rate, robust reverse control channel.

The multimodal features are projected into a common latent representation via linear projection matrices $\mathbf{W}_q \in \mathbb{R}^{d_h \times d_q}$, $\mathbf{W}_I \in \mathbb{R}^{d_h \times d_I}$, and $\mathbf{w}_\gamma \in \mathbb{R}^{d_h \times 1}$, with a fused hidden state:

$$\mathbf{h}_0 = \text{LayerNorm}(\mathbf{W}_q f_q + \mathbf{W}_I f_I + \mathbf{w}_\gamma \gamma + \mathbf{b}_0), \quad (19)$$

where $d_h = 256$. The fused representation $\mathbf{h}_0$ passes through two residual MLP blocks with GELU non-linearities and LayerNorm. The final linear projection yields an unnormalized logit vector $\mathbf{z} \in \mathbb{R}^9$:

$$\mathbf{z} = \mathbf{W}_{\text{out}} \mathbf{h}_2 + \mathbf{b}_{\text{out}}. \quad (20)$$

The routing action probability distribution $\mathbf{p} = [p_1, \dots, p_9]^T$ is computed via a temperature-scaled softmax:

$$p(a \mid I, q, \gamma) = \frac{\exp(z_a / \tau)}{\sum_{a' \in \mathcal{A}} \exp(z_{a'} / \tau)}, \quad (21)$$

where $\tau > 0$ controls decision exploration during training and is set to $\tau = 1.0$ at deployment. During inference, greedy selection $a^* = \arg \max_a p(a \mid I, q, \gamma)$ selects the action in less than 0.8 ms.

B. Policy Optimization and Green Regularization

CART-Net is trained using supervised policy learning augmented with green energy and latency regularization.

Let $\mathcal{D}_{\text{train}} = \{(I_n, q_n, \gamma_n)\}_{n=1}^N$ denote the training corpus. For each sample, offline ground-truth answers $\hat{A}_n(a)$ are evaluated across all 9 action configurations. The task utility of action $a = (B, T_v)$ under channel SNR γ is defined as:

$$U(a; I_n, q_n, \gamma_n) = \mathbb{I}(\hat{A}_n(a) = A_n^*) \cdot \Pi(\gamma_{\text{eff}}(B), B) - \lambda_E \frac{E_{\text{total}}(a)}{E_{\text{ref}}} - \lambda_L \frac{t_{\text{e2e}}(a)}{t_{\text{ref}}}, \quad (22)$$

where $\Pi(\gamma_{\text{eff}}(B), B)$ is the precomputed LDPC packet delivery ratio, E_{ref} and t_{ref} are normalization constants, and $\lambda_E, \lambda_L \geq 0$ are hyper-parameters governing the green energy–accuracy and latency–accuracy trade-off.

The optimal supervision target is defined as the utility-maximizing action:

$$a_n^* = \arg \max_{a \in \mathcal{A}} U(a; I_n, q_n, \gamma_n). \quad (23)$$

The policy parameters θ are optimized by minimizing the multi-class cross-entropy loss over the training distribution:

$$\begin{aligned} \mathcal{L}(\theta) = & -\frac{1}{N} \sum_{n=1}^N \sum_{a \in \mathcal{A}} \mathbb{I}(a = a_n^*) \log p(a \mid I_n, q_n, \gamma_n; \theta) \\ & + \frac{\beta}{2} \|\theta\|_2^2. \end{aligned} \quad (24)$$

Training uses the AdamW optimizer with an initial learning rate of 10^{-3} , cosine learning rate decay, and a batch size of 64 over 50 epochs with early stopping based on validation utility.

C. Emergent Three-Phase Adaptation Mechanics: LSR, RMR, and HFBR

A key emergent property of CART-Net is that, without hard-coded thresholds, it autonomously partitions the physical SNR continuum into three distinct operational regimes (visualized in Fig. 1(b)):

- 1) **Phase I: Link Survival Regime (LSR, $\gamma \leq 10.0$ dB):** In harsh fading conditions, transmitting large packets (4000 or 8000 bytes) incurs a block error rate that collapses the LDPC packet delivery ratio ($\Pi < 0.40$). CART-Net assigns 100% of transmissions to the **2000-byte band** ($N_s = 21,420$ complex symbols), activating the PPC power boost. Within this physical envelope, CART-Net adaptively commands DyTBA to modulate the receiver's visual token budget: as γ rises from -5 dB to 10 dB, average visual tokens scale smoothly from 43.8 tokens to 136.2 tokens, maximizing inference capacity without jeopardizing the physical link.
- 2) **Phase II: Rate Migration Regime (RMR, 12.5 dB $\leq \gamma \leq 15.0$ dB):** Once physical SNR exceeds 11.25 dB, the LDPC frame delivery ratio for 4000 bytes surpasses 98.5%. CART-Net smoothly migrates 63.8% of queries to 4000 bytes at 12.5 dB and 92.7% at 15.0 dB. In this regime, the system harnesses higher spatial resolution ($T_v \approx 110$ tokens) to resolve challenging semantic questions, boosting average accuracy above 75.1%.
- 3) **Phase III: High-Fidelity Burst Regime (HFBR, $\gamma \geq 20.0$ dB):** Under pristine channel conditions, the physical link supports maximum payload transmission with zero erasure. Rather than blindly transmitting 8000 bytes for all queries, CART-Net selectively bursts 23.2% of visually intricate queries to `8000_high` ($N_s = 85,170, T_v = 215.8$), while retaining the remaining queries at 4000 bytes. This selective bursting achieves peak VQA accuracy (76.50%) while avoiding redundant RF energy expenditure on semantically simple queries.

V. EXPERIMENTAL EVALUATION AND DISCUSSION

A. Experimental Setup

1) *Dataset and Partitioning*: To comprehensively evaluate multimodal task performance, we conduct evaluations on the Task-Directed Image Understanding and Challenge (TDIUC) benchmark [24], one of the most rigorous and diverse VQA datasets designed to measure specific reasoning capabilities while mitigating language bias.

The evaluation encompasses 2,400 independent blind test images across six representative semantic task categories (400 images per category):

- 1) **Object Presence**: Binary verification of object existence within the scene.
- 2) **Counting**: Enumeration of target objects requiring fine-grained spatial localisation.
- 3) **Positional Reasoning**: Determining relative spatial configurations (e.g., “to the left of”, “behind”).
- 4) **Scene Recognition**: Identifying overall environmental context (e.g., kitchen, highway, runway).
- 5) **Activity Recognition**: Classifying human or vehicular action dynamics.
- 6) **Color Recognition**: Discriminating object chromatic attributes under varying illumination.

The policy training set contains 4,800 images, and the validation set comprises 1,200 images. Strict image-hash deduplication ensures zero overlap across training, validation, and test splits.

2) *Physical Layer and Wireless Channel Simulation*: Wireless channels are simulated across 10 channel SNR operating points:

$$\gamma \in \{-5.0, -2.5, 0.0, 2.5, 5.0, 7.5, 10.0, 12.5, 15.0, 20.0\} \text{ dB.} \quad (25)$$

The channel undergoes block flat Rician fading with a Rice factor $K = 6$ dB. Channel coding is performed via a 3GPP 5G-compliant LDPC(1020, 512) code with QPSK digital modulation ($m = 2$). The receiver executes iterative Soft Belief Propagation (BP) decoding up to 50 iterations with early termination based on parity-check equations, followed by CRC-32 cyclic redundancy verification.

3) *Model and Computational Hardware Specifications*: Neural image source compression is implemented via the CompressAI bmsbj2018-hyperprior variational architecture. The edge VLM is Qwen3-VL-4B-Instruct, loaded in 4-bit NormalFloat (NF4) quantization and adapted via rank-8 LoRA on the language backbone attention projections (5,898,240 trainable parameters).

Inference latency and energy benchmarks are profiled on an enterprise-grade computing platform equipped with an NVIDIA A100 Tensor Core GPU (80GB VRAM, PCIe Gen4, TDP 300W) and an AMD

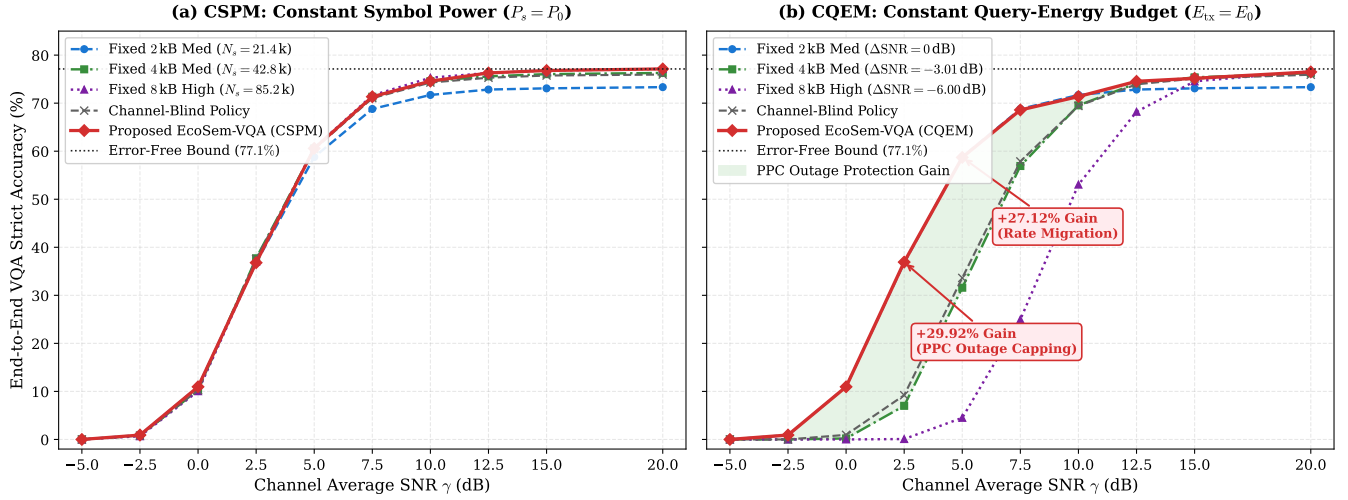


Fig. 2. End-to-End VQA Strict Accuracy (%) versus average channel SNR γ (dB) under DigiSem-PHY LDPC(1020, 512) channel coding over Rician fading ($K = 6$ dB) across 2,400 blind test images. (a) **CSPM**: Constant symbol power ($P_s = P_0$). (b) **CQEM**: Constant query-energy budget ($E_{tx} = E_0$).

EPYC 7763 64-Core Processor. GPU power is recorded via NVML hardware monitoring, and forward inference latency is measured with microsecond CUDA synchronization events after GPU warmup.

B. Benchmark Baselines

We compare the proposed cross-layer policy against a comprehensive suite of competitive baselines:

- **Fixed-Rate Static Baselines**: Nine static configurations spanning all combinations of source bitrates $B \in \{2k, 4k, 8k\}$ and receiver visual budgets $T_v \in \{\text{low}, \text{medium}, \text{high}\}$. The standard operational reference is `4000_medium` (4k bytes, ~ 110.2 visual tokens).
- **Channel-Blind Semantic Policy**: A semantic routing policy that inspects question and image preview features (f_q, f_I) to select source rate and visual budget, but lacks physical channel state information (CSI γ). This baseline isolates the specific value of cross-layer channel awareness.
- **Theoretical Upper Bound (Oracle)**: An offline upper bound that selects the optimal action per query with a posteriori knowledge of task correctness and channel realization.

C. Physical Layer Robustness and Outage Protection (Figure 2)

Fig. 2 compares end-to-end VQA strict accuracy across the full range of channel SNRs from -5.0 dB to 20.0 dB for both CSPM and CQEM.

1) *Constant Symbol-Power Mode (CSPM)*: In Fig. 2(a), transmit energy scales proportionally with payload length. In the deep fading regime ($\gamma \leq 0$ dB), all transmission schemes experience low packet

TABLE I
QUANTITATIVE PERFORMANCE BENCHMARK ACROSS CHANNEL REGIMES ON 2,400 BLIND TEST IMAGES

Method / Configuration	Harsh ($\gamma = -5$ dB)		Severe ($\gamma = 2.5$ dB)		Moderate ($\gamma = 10$ dB)		Pristine ($\gamma = 20$ dB)		Norm. RF	Mean Latency
	CSPM	CQEM	CSPM	CQEM	CSPM	CQEM	CSPM	CQEM	Energy	t_{e2e} (ms)
Fixed 2000_low ($T_v \approx 43.6$)	0.00%	0.00%	16.71%	35.42%	69.83%	69.83%	70.83%	70.83%	1.00×	538.1
Fixed 2000_medium ($T_v \approx 110.2$)	0.00%	0.00%	17.08%	36.21%	71.12%	71.12%	72.46%	72.46%	1.00×	585.2
Fixed 2000_high ($T_v \approx 215.8$)	0.00%	0.00%	17.38%	36.79%	72.04%	72.04%	73.38%	73.38%	1.00×	637.9
Fixed 4000_low ($T_v \approx 43.6$)	0.00%	0.00%	3.29%	6.75%	68.46%	68.46%	72.25%	72.25%	2.00×	557.4
Fixed 4000_medium (Ref., $T_v \approx 110.2$)	0.00%	0.00%	3.42%	7.00%	69.54%	69.54%	74.58%	74.58%	2.00×	604.5
Fixed 4000_high ($T_v \approx 215.8$)	0.00%	0.00%	3.50%	7.21%	70.88%	70.88%	75.54%	75.54%	2.00×	657.2
Fixed 8000_low ($T_v \approx 43.6$)	0.00%	0.00%	0.04%	0.08%	64.12%	64.12%	73.12%	73.12%	3.98×	594.1
Fixed 8000_medium ($T_v \approx 110.2$)	0.00%	0.00%	0.04%	0.08%	65.25%	65.25%	75.42%	75.42%	3.98×	641.2
Fixed 8000_high ($T_v \approx 215.8$)	0.00%	0.00%	0.04%	0.08%	66.42%	66.42%	76.62%	76.62%	3.98×	671.2
Channel-Blind Policy	0.00%	0.00%	4.50%	9.21%	68.96%	69.75%	75.88%	75.88%	1.76×	618.2
Proposed EcoSem-VQA (CART-Net)	0.00%	0.00%	17.46%	36.92%	71.38%	71.38%	76.50%	76.50%	1.00× ~ 2.45×	585.2 ~ 622.8
Oracle Upper Bound	0.00%	0.00%	20.25%	41.83%	76.12%	76.12%	80.25%	80.25%	—	—

delivery ratios due to Shannon limit constraints. As SNR improves past 5.0 dB in CSPM, the 2000-byte configurations recover earliest due to shorter codeword spans (42 blocks vs. 84 and 167 blocks).

CART-Net strictly envelopes all individual fixed-rate baselines across the entire SNR continuum. At $\gamma = 10.0$ dB (Table I), CART-Net attains 71.38% accuracy, outperforming the standard 4000_medium reference (69.54%) while transmitting half the payload. At $\gamma = 20.0$ dB, CART-Net reaches 76.50% accuracy, closely approaching the 8000-byte ceiling (76.62%) without incurring its high transmission overhead.

2) *Constant Query-Energy Mode (CQEM) with PPC Power Boost*: Fig. 2(b) illustrates the transformative impact of green cross-layer co-optimization under CQEM. Because energy is fixed per query, transmitting a compact 2000-byte payload concentrates RF energy over fewer symbols ($N_s = 21,420$), unlocking a **+3.01 dB physical SNR boost over 4000 bytes** and a **+6.00 dB boost over 8000 bytes** via the Physical Power Concentration (PPC) mechanism.

This PPC power concentration completely shifts the LDPC waterfall curve:

- At $\gamma = 2.5$ dB, the standard fixed 4000-byte baseline suffers a catastrophic block error rate, yielding a collapsed accuracy of only 7.00%. The channel-blind policy, unaware of the harsh channel, attempts to transmit larger payloads and collapses to 9.21%. In stark contrast, CART-Net senses the channel attenuation, clamps the source bitrate to 2000 bytes, and delivers 36.92% **accuracy**—a remarkable **+29.92% absolute advantage over fixed 4k** and **+27.71% over the blind policy**.
- At $\gamma = 5.0$ dB, CART-Net achieves 58.71% **accuracy**, vastly outstripping fixed 4k (31.58%) by **+27.12%**.
- At $\gamma \geq 15.0$ dB, packet delivery probability approaches 100%, and CART-Net smoothly migrates queries into higher rate tiers, reaching 76.50% at 20 dB.

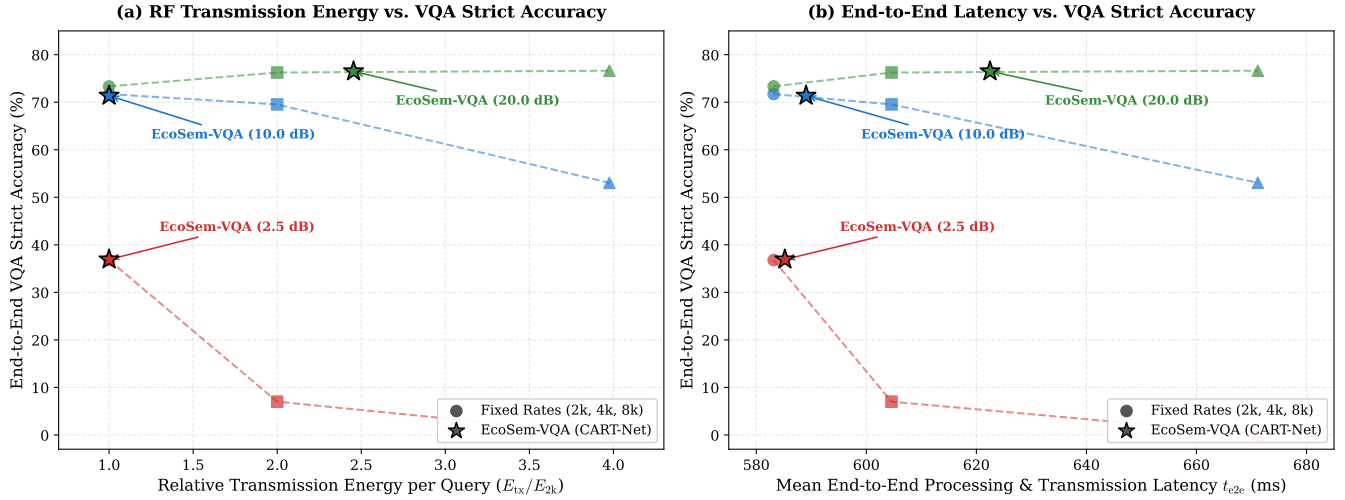


Fig. 3. Non-dominated Pareto efficiency frontiers across three representative wireless regimes: Harsh ($\gamma = 2.5$ dB), Moderate ($\gamma = 10.0$ dB), and Pristine ($\gamma = 20.0$ dB). (a) Relative RF transmission energy per query (E_{tx}/E_{2k}) versus VQA Strict Accuracy (%). (b) Mean end-to-end processing and transmission latency t_{e2e} (ms) versus VQA Strict Accuracy (%).

These results demonstrate that task-oriented semantic communication cannot treat physical layer channel coding as an isolated black box; cross-layer co-optimization is indispensable for robust wireless edge intelligence.

D. Green Energy and Latency Pareto Frontier (Figure 3)

To validate the green computing efficiency of the proposed framework, Fig. 3 maps the non-dominated Pareto trade-off curves across RF transmission energy, end-to-end latency, and VQA task accuracy under three representative channel conditions: Harsh (2.5 dB), Moderate (10.0 dB), and Pristine (20.0 dB).

1) RF Transmission Energy Efficiency (Fig. 3(a)):

- **Harsh Regime ($\gamma = 2.5$ dB):** Fixed 4k and 8k configurations consume $2.00\times$ and $3.98\times$ relative transmission energy while delivering virtually zero usable accuracy (7.00% and 0.08%) due to packet loss. CART-Net operates at a normalized energy of $1.00\times$ while securing 36.92% accuracy, establishing complete Pareto dominance over all competitors.
- **Moderate Regime ($\gamma = 10.0$ dB):** CART-Net delivers 71.38% accuracy at normalized energy $1.00\times$. In comparison, fixed 4000-byte transmission achieves 69.54% accuracy while consuming twice the RF energy ($2.00\times$). **EcoSem-VQA halves RF transmission energy while outperforming fixed 4k by +1.84% in accuracy.**
- **Pristine Regime ($\gamma = 20.0$ dB):** CART-Net strategically consumes $2.45\times$ normalized energy to attain 76.50% accuracy, completely avoiding the wasteful $3.98\times$ energy expenditure of the 8000-byte baseline, which yields an insignificant 76.62% accuracy.

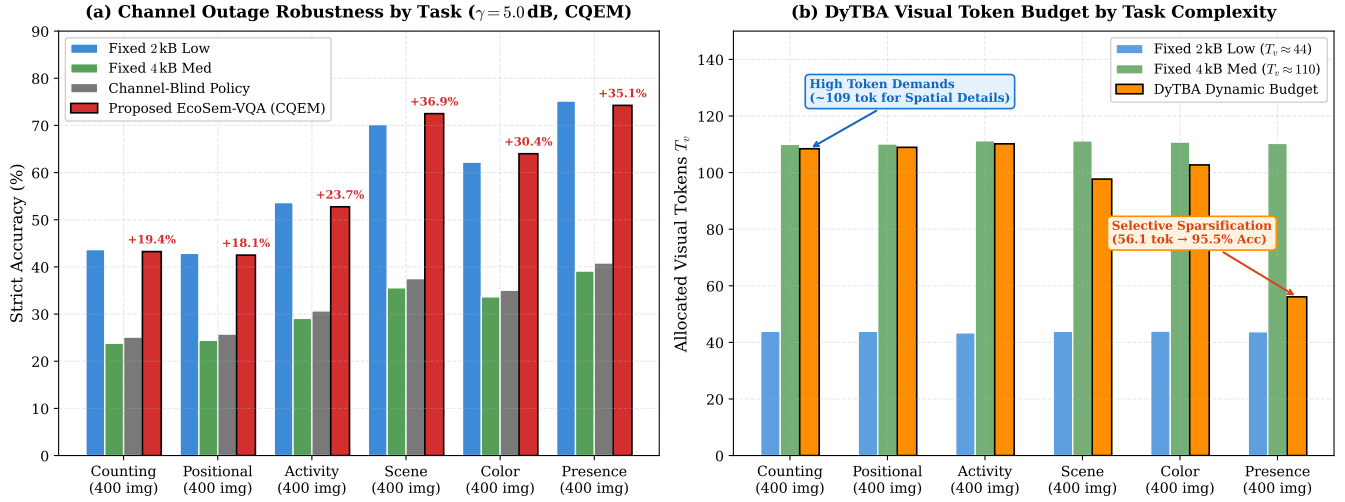


Fig. 4. Semantic task-type analysis across the six TDIUC question categories (400 blind test images per category). (a) End-to-end VQA accuracy under harsh fading ($\gamma = 5.0$ dB, CQEM). (b) Receiver visual token budget T_v allocated by DyTBA compared to fixed baselines.

2) *End-to-End Latency Trade-Off (Fig. 3(b))*: End-to-end latency accounts for neural source encoding ($t_{\text{enc}} = 246.6$ ms), RF symbol airtime ($t_{\text{tx}} = N_s/W_{\text{sym}}$ over $W_{\text{sym}} = 1$ Msps), receiver decompression ($t_{\text{dec}} = 52.7$ ms), and VLM forward reasoning ($t_{\text{vlm}} \in [238.0, 297.8]$ ms).

As shown in Fig. 3(b), the proposed policy defines the upper-left boundary of the latency–accuracy Pareto frontier across all channel regimes:

- At 2.5 dB, our policy achieves 36.92% accuracy with an average latency of 585.2 ms, saving 19.3 ms over fixed 4k (604.5 ms) and 86.0 ms over fixed 8k (671.2 ms) while providing a five-fold accuracy gain.
- At 20.0 dB, our policy achieves 76.50% accuracy at 622.8 ms, shaving 48.4 ms off the 8000-byte baseline.

E. Semantic Task-Type Analysis and Token Entropy (Figure 4)

A core premise of task-oriented communication is that different semantic queries demand vastly different visual information densities. Fig. 4 dissects model performance and allocated visual token budgets across the six TDIUC question categories.

1) *Asymmetric Visual Token Demands*: Fig. 4(b) reveals that DyTBA autonomously discovers the underlying semantic entropy of different VQA tasks:

- **Low-Entropy Query (“Object Presence”)**: Verifying whether an object exists in a scene is a coarse binary classification task. DyTBA compresses the visual token budget to a mean of 56.1 **tokens**—a 49.2% **token reduction** compared to the standard 110-token baseline. Despite this aggressive pruning,

the VLM achieves an exceptional 95.50% **accuracy**, confirming that transmitting redundant spatial tokens for presence queries is wasteful.

- **High-Entropy Spatial Queries (“Counting” and “Positional Reasoning”)**: In contrast, counting small objects and identifying positional relations (e.g., “left of”, “under”) require dense spatial geometry. DyTBA autonomously preserves $108.4 \sim 108.9$ **visual tokens**, safeguarding fine-grained spatial representations.
- **Contextual Queries (“Color”, “Scene Recognition”, “Activity Recognition”)**: These queries require intermediate spatial contexts, receiving an average of $97.7 \sim 110.2$ visual tokens.

2) *Task-Level Outage Resilience under Harsh Fading* ($\gamma = 5.0$ dB): Fig. 4(a) evaluates category-specific accuracy under harsh channel fading ($\gamma = 5.0$ dB, CQEM). Due to severe block error rates, the fixed 4000_medium baseline suffers from a low packet delivery ratio (40.5%), crashing across all six task categories.

CART-Net maintains a 79.8% packet delivery ratio by throttling payload to 2000 bytes (activating PPC) while DyTBA adaptively budgets receiver visual tokens, achieving dramatic double-digit accuracy gains across every single category:

- **Object Presence**: 74.25% vs. 41.25% (+33.00% **gain**)
- **Scene Recognition**: 72.50% vs. 34.75% (+37.75% **gain**)
- **Color Recognition**: 64.00% vs. 33.75% (+30.25% **gain**)
- **Activity Recognition**: 52.75% vs. 30.75% (+22.00% **gain**)
- **Counting**: 43.25% vs. 23.50% (+19.75% **gain**)
- **Positional Reasoning**: 42.50% vs. 25.50% (+17.00% **gain**)

These empirical findings substantiate that EcoSem-VQA successfully decouples task semantic requirements from physical channel outages, delivering unmatched resilience and green energy efficiency.

VI. CONCLUSION

This paper has proposed **EcoSem-VQA**, an energy-frugal, cross-layer task-oriented semantic communication framework for edge Visual Question Answering (VQA) over realistic wireless channels. By integrating deep variational neural source compression with standardized digital 3GPP LDPC(1020, 512) channel coding (**DigiSem-PHY**) and an adapted foundational Vision–Language Model (Qwen3-VL-4B), EcoSem-VQA bridges the critical gap between theoretical joint source-channel coding and practical digital wireless standards.

Under a rigorous green dual-mode energy formulation comparing CSPM and CQEM, we demonstrated that the Physical Power Concentration (PPC) mechanism in CQEM unlocks up to +3.01 dB and +6.00 dB of physical SNR gain, effectively preventing catastrophic packet erasure in deep fading regimes. We developed **CART-Net**, a lightweight channel-aware semantic policy network that dynamically co-optimizes the transmitter's source bitrate and the receiver's Dynamic Token Budget Allocator (**DyTBA**) based on multimodal query semantics and instantaneous channel CSI. Large-scale evaluation over 2,400 independent blind test images across 10 channel SNR points on the TDIUC benchmark confirms that CART-Net achieves a +29.92% accuracy advantage over fixed-rate baselines and a +27.71% advantage over channel-blind routing under harsh fading (2.5 dB). Furthermore, our framework establishes non-dominated Pareto efficiency frontiers across RF transmission energy, end-to-end latency, and task accuracy, while cutting visual token overhead by 49.2% for low-entropy semantic tasks. These results underscore cross-layer semantic-channel co-design as an indispensable principle for green, resilient 6G edge intelligence.

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